Abdullah Al Zahid

Toronto, ON | (780) 982-3680 | azahid1@ualberta.ca | linkedin.com/in/abdullah-al-zahid

SUMMARY

Transportation engineer with a master's degree, currently an Assistant Planner with the City of Toronto, where GIS and Python are the tools behind much of the work. Builds and maintains large spatial datasets, a validated data dashboard and tooling that automates analysis. Earlier GIS work covered wildfire evacuation planning in Alberta communities.

EXPERIENCE

Assistant Planner, Transportation Planning

City of Toronto | January 2024 – present

- Built and maintain the road and right-of-way datasets Transportation Planning depends on, including a right-of-way width dataset for all city roads, using QGIS, MapBox, JavaScript and Python to merge sources that were scattered across the division.
- Built the Transportation Data Dashboard, replacing a report rebuilt by hand each quarter with one validated source the division reports from, and saw the change through to adoption.
- Analyze travel and network data, including several years of the Transportation Tomorrow Survey, to establish where capacity falls short and what that means for future service and investment.
- Analyze large volumes of travel and network data for trends, gaps and emerging issues, and turn them into KPIs and data-driven recommendations.
- Run assigned projects from scoping through to recommendations, planning the work, holding the schedule and quality myself, and reporting progress to the people relying on it.
- Write the briefing notes, memos and presentations that carry issues, options and recommendations to senior staff across divisions.

Transportation Planner, Simulation and Modelling

Metrolinx, Service Design | February 2023 – January 2024

- Built the Crew Variance Tool, a Python application that read large JSON operations files and produced summaries of speed profiles and run-time variance across subdivisions.
- Set up a command line workflow so temporary and permanent slow orders could be pre-simulated in batches, cutting turnaround on scenario requests from days to hours.
- Ran OpenTrack simulations of GO rail services to establish what proposed operating and infrastructure changes would do to run times, reliability and recovery before they were taken forward.

Graduate Transportation Planner

Mott MacDonald | January – February 2023

- Short consulting placement on Ontario Line work. Ran traffic simulations to estimate how construction staging would affect surrounding streets, and prepared the supporting material for the project team.

Graduate Research Assistant

University of Alberta | January 2020 – August 2022

- Master's research on how road networks perform during emergency evacuations, using macroscopic (LWR) and agent-based (MATSim) models.
- Worked with large travel datasets and wrote up results for readers without a modelling background.

SKILLS

ArcGIS Pro, QGIS and MapBox. Python, SQL and JavaScript. Spatial analysis, data validation and dashboards. Large multi-source datasets. AutoCAD and Civil 3D. Synchro, VISSIM, OpenTrack and MATSim. MS Office.

EDUCATION

MSc, Transportation Engineering — University of Alberta, 2022

BSc, Civil Engineering — Bangladesh University of Engineering and Technology, 2018

BSc, Computer Science — University of British Columbia, in progress (2025–2027)